

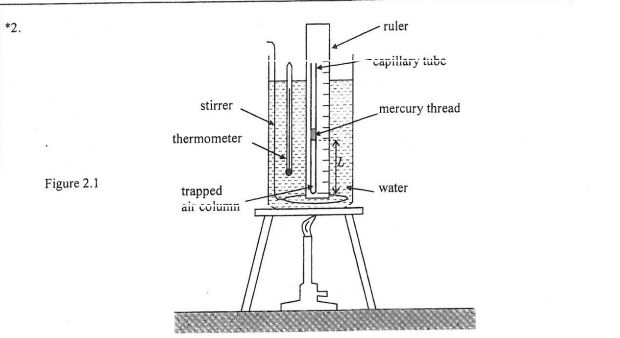


Figure 2.1 shows an air column trapped by a small mercury thread inside a uniform capillary tube. The set-up is heated by a water bath. The length of the air column L is measured at various temperature θ . Some of the results are tabulated below:

Temperature $\theta / ^\circ\text{C}$	20	92
Length of air column L / mm	64	80

- (a) Describe the procedure(s) to be done before taking a reading in order to ensure that the trapped air reaches the same temperature as the water. (2 mark)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Please stick the barcode label

- (b) Assume that length L increases linearly with temperature θ throughout.
- (i) Estimate the length of the air column when the temperature indicated by the thermometer is 65°C . (2 marks)

- (ii) Find the 'absolute zero' as obtained from this experiment. (2 marks)

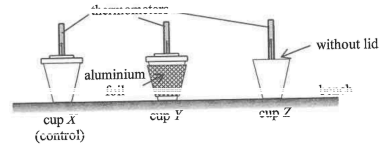
Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

1. A student conducts an experiment to study how to best retain the warmth of water by using different types of cups. Each cup is filled with 250 cm³ of hot water and cup X serves as a control.

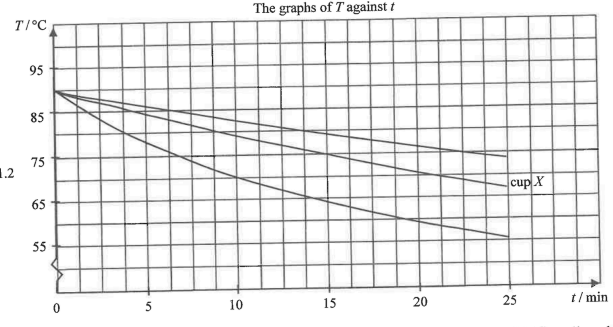
Figure 1.1



cup	wrapping	with a lid
X	no	yes
Y	aluminium	yes
Z	no	no

When the water temperature is 90 °C, the student starts taking thermometer readings every minute. Figure 1.2 shows the variation of temperature (T) of water in the cups with time (t) elapsed.

Figure 1.2



(a) Suggest a reason why this experiment begins with the same initial water temperature (90 °C). (1 mark)

Answers written in the margins will not be marked.

Please stick the barcode label here

(b) Explain why all the curves become less steep as time elapses. (2 marks)

(c) (i) On Figure 1.2, indicate the respective curves that correspond to the results of cup Y and cup Z. (1 mark)

(ii) Explain your answer by referring respectively to the main heat transfer process. (3 marks)

(d) Suggest a material for making cups so that heat loss via conduction can be reduced. (1 mark)

Answers written in the margins will not be marked.